

FE-APP-05 - Reporting Portal

Version: v1.0 **Bundle:** 99_frontend **Status:** Draft for review **Primary owner:** Frontend / reporting product team **Related DDs:** FE-APP-00, FE-APP-01, DD_REP-01, DD_EM-CFG-03, DD_ILM-CFG-01, DD_BIL-01, DD_BIL-04, DD_SUB-LM-01, DD_WO-01, DD_TCK-01, DD_EM-03, DD_OSR-01, DD_FA-01, DD_FA-02, DD_FA-03

MVP simplification baseline: first release reporting portal shows only core operational/revenue dashboards and async exports backed by REP-01. It must not call operational module APIs for aggregate totals. Drill-down may link to source modules for a specific entity only.

1. Purpose

FE-APP-05 defines the management reporting user experience for SOPHIX BSS.

The portal gives managers, supervisors, finance, commercial operations, network operations, field operations, and executives a controlled way to view dashboards, drill into KPIs, schedule exports, and inspect reconciliation exceptions.

The portal is Kenya/WIK-first but must remain operator-configurable for other regions: language, currency, timezone, brand, KPI labels, dashboard visibility, fiscal calendar, export formats, and data access rules.

FE-APP-05 consumes REP - 01 dashboard/export APIs. It does not query operational module databases directly.

2. Application Placement

For a budget-conscious medium deployment, start FE-APP-05 inside the Backoffice shell as a Reporting area. Split it into a separate portal only when reporting users, permission scope, dashboard volume, or export scheduling justify a separate deployment.

Option	Recommendation
Start as Backoffice module	Preferred for initial Kenya launch.
Separate reporting portal	Use when executive/partner users need isolated access.
Direct BI tool only	Not recommended as the sole UX; SOPHIX still needs controlled operational drilldowns and export governance.

3. Scope And Boundaries

Area	Included features	Primary module APIs
Executive dashboards	Revenue, subscribers, churn, installations, tickets, stock, dunning.	REP - 01 dashboard APIs.
Finance reporting	Invoices, payments, aging, dunning, revenue summaries, exports.	REP - 01, sourced from BIL modules.
Commercial reporting	Sales funnel, package mix, discounts, campaigns, CVM outcomes.	REP - 01, SALES - 01, SIP, EM - 03.
Operations reporting	WO SLA, installation backlog, support jobs, contractor performance.	REP - 01, sourced from WO/TCK/ASR.
Network reporting	HomePass readiness, audit findings, migration success, provisioning reconciliation.	REP - 01, RLM, FUL, FA-02.
Inventory reporting	Stock balances, procurement, RMA, equipment audits.	REP - 01, OSR modules, FA-01.
Export center	CSV/XLSX/PDF exports, scheduled reports, download history.	REP - 01 export APIs.
Drilldown	Read-only link back to Backoffice records.	FE-APP-01 routes + owning module read APIs.

Boundary rules:

- FE-APP-05 does not compute authoritative KPIs in browser code.
- FE-APP-05 does not call operational write APIs.
- FE-APP-05 must not bypass REP-01 permissions or row-level filters.

- Drilldowns must preserve module ownership: ticket details from TCK, WO details from WO, invoice details from BIL, customer details from ILM.

4. Users And Roles

Roles come from EM-CFG-03/RBAC.

User	Main needs	Key APIs
Executive	High-level KPIs, trend, exceptions.	GET /api/reporting/dashboards/executive.
Finance manager	Revenue, aging, payments, exports.	REP-01 finance datasets/exports.
Commercial manager	Sales, package, discounts, CVM.	REP-01 commercial dashboards.
Operations supervisor	WO/ticket SLA and backlog.	REP-01 operations dashboards.
Network manager	HomePass, provisioning, audit.	REP-01 network dashboards.
Inventory manager	Stock, procurement, RMA, audit.	REP-01 inventory dashboards.
Read-only auditor	Filtered reports and export history.	REP-01 with audit role.

5. Information Architecture

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flowchart TB
    A["Reporting Home"] --> B["Executive"]
    A --> C["Finance"]
    A --> D["Commercial"]
    A --> E["Operations"]
    A --> F["Network"]
    A --> G["Inventory"]
    A --> H["Audit And Compliance"]
    A --> I["Export Center"]
    B --> J["KPI drilldown"]
    C --> J
    D --> J
    E --> J
    F --> J
    G --> J
  
```

6. Screen Groups

6.1 Reporting Home

Purpose: show the user's allowed dashboard groups, recent exports, watched KPIs, and active data freshness warnings.

Module APIs:

UI Area	Module	Endpoint
Dashboard catalog	REP-01	GET /api/reporting/dashboard-catalog
User favorites	REP-01 or user preferences foundation	GET /api/reporting/favorites
Data freshness	REP-01	GET /api/reporting/freshness

6.2 Dashboard View

Purpose: render KPI tiles, charts, tables, filters, and drilldown actions.

Module APIs:

UI Area	Module	Endpoint
Dashboard metadata	REP-01	GET /api/reporting/dashboards/{dashboard_code}
Dashboard data	REP-01	POST /api/reporting/dashboards/{dashboard_code}/query
Drilldown data	REP-01	POST /api/reporting/dashboards/{dashboard_code}/drilldown

6.3 Export Center

Purpose: request, schedule, download, cancel, and audit report exports.

Module APIs:

UI Area	Module	Endpoint
Create export	REP-01	POST /api/reporting/exports
Export list	REP-01	GET /api/reporting/exports
Download export	REP-01	GET /api/reporting/exports/{export_id}/download
Schedule export	REP-01	POST /api/reporting/export-schedules

6.4 Reconciliation And Exception Views

Purpose: show finance/provisioning/inventory/ticketing exceptions that need operational follow-up.

Module APIs:

UI Area	Module	Endpoint
Reconciliation summary	REP-01	GET /api/reporting/reconciliation/summary
Exception list	REP-01	POST /api/reporting/reconciliation/exceptions/search
Source record drilldown	Owning module	BIL/TCK/WO/OSR/FUL read APIs

7. Dashboard Catalog For V1

Dashboard	Purpose	Key sources through REP-01
EXEC_OVERVIEW	Company-level KPI overview.	BIL, SUB, TCK, WO, OSR.
FIN_REVENUE	Revenue, invoice, payment, aging.	BIL-01/BIL-02/BIL-04.
FIN_DUNNING	Dunning levels, recovery, restriction/suspension trend.	BIL-04, SUB-WF.
COM_SALES_FUNNEL	Leads, conversion, package mix.	SALES, FUL-02, SIP.
COM_CVM	CVM activity, retention, win-back outcomes.	EM-03, CUST-INT.
OPS_WO_SLA	Installation/support/shifting WO SLA.	WO-01.
OPS_TICKETS	Tickets, complaints, SLA, backlog.	TCK/ASR.
NET_HOMEPASS	HomePass readiness and sellability.	RLM, FA-02.
NET_PROVISIONING	Activation/cutover/reconciliation failures.	FUL/PROV-INT.
INV_STOCK	Stock, procurement, RMA, audit.	OSR, FA-01.

8. UX Requirements

- Use dense dashboard layouts, not marketing-style cards.
- Every dashboard must show data freshness and filter scope.
- Every KPI with a number must have a clear definition tooltip.
- Every chart must support table view for auditability.
- Export buttons must show permission and row-limit warnings.
- Drilldown links must say which module owns the record.
- Empty states must distinguish no data, no permission, and data delay.

9. Security

- Enforce RBAC through REP-01 and the frontend route guard.
- Never expose raw KYC documents, full identity numbers, payment secrets, or internal credentials.
- Apply row-level filters by operator, franchise, region, team, and role.
- Export actions must be audited.
- Downloads should expire and use signed URLs or controlled file download APIs.

10. Configuration

Config	Purpose
enabledDashboardCodes	Dashboards enabled for operator.
defaultCurrency	KES for Kenya first deployment.
timezone	Africa/Nairobi for WIK if configured.
fiscalCalendar	Month/quarter definitions.
brandTheme	Operator logo/colors.
languagePacks	English first, extensible to other languages.
exportFormats	CSV/XLSX/PDF availability.
rowLevelFilterMode	Operator/franchise/team restrictions.

11. Test Scenarios

1. User sees only allowed dashboards.
2. Dashboard data is loaded only from REP-01.
3. Finance export respects row-level filters.
4. Drilldown opens Backoffice record with owning module route.
5. Data freshness warning appears when mart load is stale.
6. No permission and no data render different states.
7. Export download is audited.